

The logic of arguments - Set 1

TMUA practice questions by topic. Specification Arg1-Arg4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Consider the statement: **if n is a multiple of 6, then n is a multiple of 3**. Which of the following is its contrapositive?
- A If n is not a multiple of 3, then n is not a multiple of 6.
 - B If n is a multiple of 3, then n is a multiple of 6.
 - C If n is not a multiple of 6, then n is not a multiple of 3.
 - D n is a multiple of 6 if and only if n is a multiple of 3.
 - E If n is a multiple of 3, then n is not a multiple of 6.
- 2 Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .
- 3 Which of the following is the negation of the statement '**every student in the class scored more than 50**'?
- A At least one student in the class scored more than 50.
 - B Every student in the class scored 50 or less.
 - C Every student in the class scored less than 50.
 - D At least one student in the class scored 50 or less.
 - E No student in the class scored more than 50.

- 4 Consider the statement about a real number x : **if** $x^2 = x$, **then** $x = 1$. Which one of the following is true?
- A The statement is true and its converse is true.
 - B The statement is true and its converse is false.
 - C The statement is false and its converse is true.
 - D The statement is false and its converse is false.
- 5 Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .
- 6 Which of the following is the negation of '**there exists a positive integer n such that $n^2 + n + 41$ is not prime**'?
- A For every positive integer n , $n^2 + n + 41$ is not prime.
 - B There exists a positive integer n such that $n^2 + n + 41$ is prime.
 - C For every positive integer n , $n^2 + n + 41$ is prime.
 - D For some positive integer n , $n^2 + n + 41$ is prime.
 - E There is no positive integer n for which $n^2 + n + 41$ is prime.
- 7 For real numbers a and b , let P : ' $a = b$ ' and Q : ' $a^2 = b^2$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 8 The statement '**if a triangle is equilateral then it is isosceles**' is true. Which of the following must therefore also be true?
- I If a triangle is isosceles then it is equilateral.
 - II If a triangle is not isosceles then it is not equilateral.
 - III If a triangle is not equilateral then it is not isosceles.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 9 Which of the following is the negation of ' $x > 3$ **and** $y \leq 5$ '?
- A $x < 3$ and $y > 5$
- B $x < 3$ or $y \geq 5$
- C $x \leq 3$ or $y \geq 5$
- D $x \leq 3$ and $y > 5$
- E $x \leq 3$ or $y > 5$
- 10 For a positive integer n , consider the statement: **if n is prime, then n is odd**. Which one of the following is true?
- A The statement is true and its converse is true.
- B The statement is true and its converse is false.
- C The statement is false and its converse is true.
- D The statement is false and its converse is false.

- 11 Let x be a real number. Which of the following statements is equivalent to ' $x \geq 2$ if and only if $x^3 \geq 8$ '?
- A $x^3 \geq 8 \Rightarrow x \geq 2$
 - B $(x \geq 2 \Rightarrow x^3 \geq 8)$ and $(x^3 \geq 8 \Rightarrow x \geq 2)$
 - C $(x < 2 \Rightarrow x^3 \geq 8)$ or $(x^3 < 8 \Rightarrow x \geq 2)$
 - D $x \geq 2 \Rightarrow x^3 \geq 8$
 - E $x \geq 2$ or $x^3 \geq 8$
- 12 Let f be a function defined for all real numbers. P : ' $f'(a) = 0$ '; Q : ' f has a stationary point at $x = a$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .